

4 (short)

A flexible, inextensible cable is suspended between points A and B as shown in Fig. 4. The cable is subject to a load per unit horizontal length that varies linearly from 2 kN m^{-1} to 0 kN m^{-1} . At support B, the inclination of the cable is 14° to the horizontal. What is the sag of the cable at midspan?

[10]

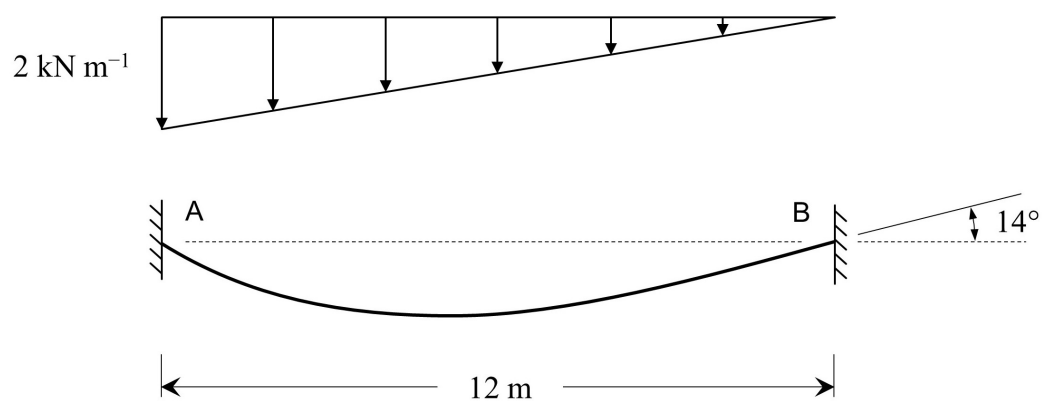


Fig. 4